

# Water–Thermal Management of FCEVs Based on Adaptive Equivalent Consumption Minimization Strategy

Geunyoung Park<sup>1</sup> and Kyunghwan Choi<sup>2</sup>

**Abstract**—Thermal management of fuel cell electric vehicles must minimize auxiliary energy consumption while keeping component temperatures within allowable ranges, yet the temperature flexibility of the stack is limited by membrane hydration. This study proposes a water–thermal adaptive equivalent consumption minimization strategy (WT-AECMS) that coordinates system-level heat allocation using requested heat-transfer rates as control inputs. Stack water dynamics determine hydration-dependent temperature bounds, and adaptive costates account for the component temperatures relative to their admissible ranges. The strategy is evaluated through high-fidelity GT-SUITE/Python co-simulation over the WLTC at an ambient temperature of  $-10^{\circ}\text{C}$ . WT-AECMS reduces the energy consumption of the vehicle thermal management system by approximately 61% relative to the baseline controller while improving thermal and hydration constraint satisfaction. Compared with thermal-only AECMS, it increases the allowable hydration time ratio from 0.70 to 1.00 with approximately 2% additional energy consumption, indicating that membrane hydration can be maintained while preserving most of the energy savings of thermal optimization.

## I. INTRODUCTION

### A. Background

The vehicle thermal management system (VTMS) of a fuel cell electric vehicle (FCEV) must keep the key components, such as the fuel cell stack, motor, and power electronics (PE), within their allowable temperature ranges to ensure performance and durability, while minimizing its own electrical consumption, which directly reduces vehicle energy efficiency [1], [2].

This task becomes challenging at low ambient temperatures, where component temperature regulation and cabin heating compete for limited waste heat, and any deficit must be covered by auxiliary electric heaters [3]. Effective waste heat recovery and allocation are therefore essential for reducing auxiliary heating power [4]. Moreover, since the allowable ranges do not prescribe a single operating temperature, the component thermal masses can store waste heat for later use, so heat allocation must be decided not only across components but also over time.

However, the temperature flexibility of the stack is limited by its membrane hydration state, as temperature and hydration jointly affect stack performance and durability [5], [6]. A higher stack temperature raises the saturation vapor pressure and promotes membrane drying, which reduces

proton conductivity and may accelerate degradation, whereas a lower temperature promotes condensation, and the resulting water accumulation impedes reactant gas transport [7], [8].

This coupling directly affects heat allocation: retaining waste heat in the stack promotes drying, whereas extracting it for cabin heating or rejecting it promotes flooding. The allowable stack temperature range should therefore be adapted to the stack current and the present hydration state, since fixed limits may permit temperatures at which the membrane dries out or floods. Effective VTMS control therefore requires allocating waste heat across components and over time to minimize electrical consumption and satisfy the cabin heating demand, while keeping the stack within hydration-dependent temperature limits.

### B. Literature Review

Previous VTMS control studies commonly formulate control decisions in terms of actuator commands, such as pump and compressor speeds and valve openings [9], [10]. In integrated thermal management systems, however, each actuator influences multiple components through shared coolant circuits and heat exchangers, so the resulting heat flows depend jointly on the combined actuator operation and the current thermal conditions [11]. This nonlinear, state-dependent relationship complicates optimization [12] and leaves the allocation of heat among the components, the cabin, and the ambient only implicit, obscuring the trade-offs among competing thermal demands.

Water management has also been integrated into energy management strategies (EMSs), as membrane hydration affects stack output while the operating current governs water generation and transport. [13] incorporated stack water content into a health-conscious EMS, and [14] coordinated power allocation with a current-dependent purge strategy to improve voltage stability and reduce hydrogen consumption. These studies, however, address power allocation between the power sources rather than heat allocation in the VTMS, and the latter explicitly identified thermal-management effects as future work.

Other studies have jointly addressed thermal and water management of the fuel cell stack. [15] coordinated stack temperature and membrane water content using model predictive control (MPC), and [16] developed a multi-input multi-output fuzzy controller for temperature and humidity regulation, both tracking prescribed operating targets. [17] relaxed these targets into operating zones through zone economic MPC, accounting for auxiliary power to improve net power output. However, these studies remain confined

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to the fuel cell subsystem, where the decisions are limited to the stack's own cooling and humidification and waste heat is treated only as a load to be rejected. At the vehicle level, the decision space expands considerably: waste heat can be retained in the component thermal masses, recovered for cabin heating, or rejected to the ambient, and each choice changes the auxiliary heating demand. Subsystem-level formulations cannot capture this trade-off.

In summary, actuator-level formulations leave heat allocation implicit, water-aware EMSs address power rather than heat, and joint thermal–water controllers remain at the subsystem level. To address these gaps, this paper formulates VTMS control as a water-constrained heat-allocation problem with requested heat-transfer rates as control inputs, which a lower-level allocation maps to actuator commands at minimum electrical power. The formulation minimizes VTMS energy consumption while satisfying the cabin heating demand and component temperature limits, with hydration-dependent stack bounds derived from the stack water dynamics.

Building on this formulation, this paper develops a water–thermal extension of the equivalent consumption minimization strategy (ECMS) used in hybrid-vehicle energy management [18], termed WT-ECMS. Applying Pontryagin's minimum principle (PMP) yields a Hamiltonian that assigns an equivalent power cost to changes in stored thermal energy, with the component temperatures playing the role of the battery state of charge. The costates are further adapted according to how close each temperature is to its admissible bounds, including the hydration-dependent stack bounds, yielding an adaptive WT-ECMS (WT-AECMS) [19]. Consequently, the decision to retain, recover, or reject heat is made at each control interval by weighing immediate electrical consumption against the value of stored heat, while keeping the stack within its hydration-dependent limits.

### C. Contributions

The main contributions of this study are as follows:

- A system-level VTMS formulation with heat-transfer rates as control inputs and hydration-dependent stack temperature bounds that maintain the membrane water content within its allowable range.
- WT-AECMS, which allocates heat by instantaneous Hamiltonian minimization with costates adapted to the component temperatures relative to their admissible ranges.

### D. Organization

The remainder of this paper is organized as follows. Section II formulates the control problem, and Section III introduces the proposed water–thermal management strategy. Section IV evaluates its performance through simulations. Section V summarizes the findings and outlines future work.

## II. PROBLEM FORMULATION

### A. Thermal Management System Model

The temperature dynamics of each component account for waste heat  $\dot{Q}_{\text{gen},i}$  generated according to its operating

efficiency, heat supplied by the heater  $\dot{Q}_{\text{heater},i}$ , heat exchange through the cooling loops  $\dot{Q}_{\text{HX},i}$ , and heat exchange with the ambient air  $\dot{Q}_{\text{amb},i}$ , as follows:

$$\dot{T}_i = \frac{1}{C_i} \left( \dot{Q}_{\text{gen},i} + \dot{Q}_{\text{heater},i} - \dot{Q}_{\text{HX},i} - \dot{Q}_{\text{amb},i} \right), \quad (1a)$$

$$\dot{Q}_{\text{heater},i} = \eta_{\text{heater},i} \cdot P_{\text{heater},i}, \quad (1b)$$

$$\dot{Q}_{\text{HX},i} = U(\dot{m}_i, \dot{m}_a) A_{\text{HX}} (T_i - T_a), \quad (1c)$$

$$\dot{Q}_{\text{amb},i} = h_{\text{amb},i} (v_{\text{veh}}) A_{\text{amb},i} (T_i - T_{\text{amb}}), \quad (1d)$$

where the subscript  $i$  identifies a component;  $T_i$  and  $C_i$  denote its representative temperature and thermal capacitance, respectively;  $\eta_{\text{heater},i}$  and  $P_{\text{heater},i}$  are the heater efficiency and electrical power input;  $\dot{m}$  denotes the fluid mass flow rate, and the subscript  $a$  identifies the other fluid stream (coolant, refrigerant, or air) exchanging heat with component  $i$  through the HX;  $U$  and  $A_{\text{HX}}$  are the overall heat transfer coefficient and heat transfer area of the HX, respectively;  $v_{\text{veh}}$  is the vehicle speed; and  $h_{\text{amb},i}$  and  $A_{\text{amb},i}$  are the ambient heat transfer coefficient and exposed area of component  $i$ , respectively.

### B. Water Management System Model

Water is generated at the cathode, transported through the inlet and outlet flows, and exchanged between the gas channels and the membrane. Within the membrane, diffusion is driven by the water-content difference at each channel interface, while electro-osmotic drag transports water from the anode toward the cathode. Accordingly, the membrane water content and channel water inventories evolve as follows:

$$\dot{\lambda}_m = \frac{\dot{n}_{w,\text{An} \rightarrow m} - \dot{n}_{w,m \rightarrow \text{Ca}}}{n_{\text{SO}_3^-}}, \quad (2a)$$

$$\dot{n}_{w,\text{An}} = \dot{n}_{w,\text{An},\text{in}} - \dot{n}_{w,\text{An},\text{out}} - \dot{n}_{w,\text{An} \rightarrow m}, \quad (2b)$$

$$\dot{n}_{w,\text{Ca}} = \dot{n}_{w,\text{gen}} + \dot{n}_{w,\text{Ca},\text{in}} - \dot{n}_{w,\text{Ca},\text{out}} + \dot{n}_{w,m \rightarrow \text{Ca}}, \quad (2c)$$

$$\dot{n}_{w,\text{An} \rightarrow m} = 2A_{\text{mem}}G_w(\lambda_{\text{eq},\text{An}} - \lambda_m) + \dot{n}_{w,\text{EOD}}, \quad (2d)$$

$$\dot{n}_{w,m \rightarrow \text{Ca}} = 2A_{\text{mem}}G_w(\lambda_m - \lambda_{\text{eq},\text{Ca}}) + \dot{n}_{w,\text{EOD}}, \quad (2e)$$

$$\lambda_{\text{eq},j} = \Lambda(a_{\text{CL},j}), \quad (2f)$$

$$a_{\text{CL},j} = \Phi_j(n_{w,j}, \lambda_m, T_{\text{st}}, i_{\text{st}}, P_j), \quad (2g)$$

where  $\lambda_m$  is the membrane water content;  $\dot{n}_{w,\text{An} \rightarrow m}$  and  $\dot{n}_{w,m \rightarrow \text{Ca}}$  are the water transfer rates from the anode to the membrane and from the membrane to the cathode, respectively;  $n_{\text{SO}_3^-}$  is the amount of sulfonic acid sites in the membrane; the subscript  $j \in \{\text{An}, \text{Ca}\}$  denotes the anode or cathode;  $n_{w,j}$  is the water inventory in channel  $j$ ;  $\dot{n}_{w,j,\text{in}}$  and  $\dot{n}_{w,j,\text{out}}$  are the inlet and outlet water flow rates;  $\dot{n}_{w,\text{gen}} = N_{\text{cell}}i_{\text{st}}/(2F)$  is the cathode water generation rate, with  $i_{\text{st}}$ ,  $N_{\text{cell}}$ , and  $F$  denoting the stack current, the number of cells, and the Faraday constant;  $A_{\text{mem}}$  is the membrane active area;  $G_w$  is the diffusive water transport coefficient;  $\dot{n}_{w,\text{EOD}}$  is the electro-osmotic water transport rate from the anode to the cathode;  $\lambda_{\text{eq},j}$  is the equilibrium water content at the membrane interface with channel  $j$ , given by the membrane sorption isotherm  $\Lambda$ ; and  $a_{\text{CL},j}$  is the catalyst-layer water activity, obtained from the map  $\Phi_j$ , with  $P_j$  denoting the gas pressure in channel  $j$ .

### C. Optimal Control Problem Formulation

The stack and PE are the thermal components retained as states. The motor is cooled by an oil circuit coupled to the PE loop through a coolant–oil HX; its heat transfer to the PE loop, governed by the motor’s internal control logic, is treated as an exogenous heat input. The state, disturbance, and control vectors are defined as

$$x(t) = [T_{\text{st}}(t), T_{\text{PE}}(t)]^T, \quad (3a)$$

$$d(t) = [v_{\text{veh}}(t), T_{\text{amb}}(t), T_{\text{cab}}(t), \dot{Q}_{\text{in}}^T(t), d_{\text{WM}}^T(t)]^T, \quad (3b)$$

$$u(t) = [\dot{Q}_{\text{rad,st}}(t), \dot{Q}_{\text{HC,st}}(t), \dot{Q}_{\text{rad,PE}}(t)]^T, \quad (3c)$$

where the subscripts st and PE denote the stack and PE, respectively;  $T_{\text{amb}}$  and  $T_{\text{cab}}$  are the ambient and cabin temperatures;  $\dot{Q}_{\text{in}} = [\dot{Q}_{\text{gen,st}}, \dot{Q}_{\text{gen,PE}}, \dot{Q}_{\text{HX,mot,PE}}]^T$  collects the heat inputs to the thermal states, including heat transferred from the motor to the PE cooling loop;  $d_{\text{WM}} = [i_{\text{st}}, n_{w,\text{An}}, n_{w,\text{Ca}}, \lambda_{\text{m}}]^T$  collects the stack current, channel water inventories, and membrane water content used in the water-management model; and  $\dot{Q}_{\text{rad,st}}$ ,  $\dot{Q}_{\text{HC,st}}$ , and  $\dot{Q}_{\text{rad,PE}}$  are the heat-transfer rates through the stack radiator, stack heater core, and PE radiator, respectively. Note that the channel pressures  $P$  and inlet and outlet flow rates in (2g) are prescribed as functions of  $i_{\text{st}}$  through predefined operating maps. Although the water inventories and membrane water content evolve according to (2), their current values are treated as given quantities in each instantaneous heat-allocation problem. They are therefore included in  $d_{\text{WM}}$  and held constant over the control interval.

The optimal control problem is formulated at the system level, where the control input specifies the required heat-transfer rates. These heat-transfer rates are realized by the lower-level VTMS actuators, whose commands are defined as

$$u_{\text{act}}(t) = \begin{bmatrix} N_{\text{fan,fr}}, N_{\text{blw,cab}}, N_{\text{comp}}, N_{\text{pump}}, \\ P_{\text{CoH}}, \theta_{\text{EV}}, \theta_{\text{flap}} \end{bmatrix}^T, \quad (4)$$

where  $N_{\text{fan,fr}}$ ,  $N_{\text{blw,cab}}$ ,  $N_{\text{comp}}$ , and  $N_{\text{pump}}$  denote the operating speeds of the front fan, cabin blower, compressor, and coolant pumps, respectively; and  $P_{\text{CoH}}$ ,  $\theta_{\text{EV}}$ , and  $\theta_{\text{flap}}$  denote the coolant-heater power, expansion-valve openings, and air-flap positions, respectively. The total VTMS electrical power consumption is defined as

$$P_{\text{VTMS}} = \sum P_{\text{heater}} + \sum P_{\text{pump}} + \sum P_{\text{fan}} + \sum P_{\text{comp}}, \quad (5)$$

where  $P_{\text{heater}}$ ,  $P_{\text{pump}}$ ,  $P_{\text{fan}}$ , and  $P_{\text{comp}}$  denote the electrical power consumption of the heating, coolant-circulation, air-flow, and compression devices, respectively. The heating devices include dedicated component heaters and the cabin PTC heater, while the airflow devices include fans and blowers. For a given heat-transfer command  $u(t)$ , the actuator command is determined by minimizing the VTMS power

consumption as

$$u_{\text{act}}^*(t) = \arg \min_{u_{\text{act}}} P_{\text{VTMS}}(u_{\text{act}}) \quad (6)$$

subject to  $\|q(x, d, u_{\text{act}}) - u(t)\| \leq \epsilon,$

where  $q(\cdot)$  maps the actuator command to the resulting heat-transfer rates, and  $\epsilon$  is the allowable tolerance. Accordingly,  $P_{\text{VTMS}}(u)$  represents the minimum power required to realize the heat-transfer command  $u$ . The cabin heating duty is determined by

$$\dot{Q}_{\text{cabDuty}} = f_{\text{cabDuty}}(T_{\text{cab}}, v_{\text{veh}}, T_{\text{amb}}), \quad (7)$$

where  $f_{\text{cabDuty}}(\cdot)$  determines the cabin heating demand from the cabin temperature, vehicle velocity, and ambient temperature.

Thermal management must also maintain the membrane water content within its allowable range,  $\lambda_{\text{m}} \leq \lambda_{\text{m}}(t) \leq \bar{\lambda}_{\text{m}}$ . Since a higher stack temperature promotes membrane drying and a lower temperature promotes hydration, this requirement is translated into stack temperature bounds. Using a one-step prediction over the control interval  $\Delta t$  with  $d_{\text{WM}}$  held constant, the set of admissible stack temperatures is

$$\mathcal{T}_{\text{WM}}(t) = \{T \in \mathcal{D}_T : \lambda_{\text{m}} \leq \lambda_{\text{m}}(t) + \Delta t f_{\lambda}(T, d_{\text{WM}}(t)) \leq \bar{\lambda}_{\text{m}}\}, \quad (8)$$

and the water-management temperature bounds are defined as

$$\begin{aligned} \underline{T}_{\text{st,WM}}(d_{\text{WM}}(t)) &= \min \mathcal{T}_{\text{WM}}(t), \\ \bar{T}_{\text{st,WM}}(d_{\text{WM}}(t)) &= \max \mathcal{T}_{\text{WM}}(t), \end{aligned} \quad (9)$$

where  $\mathcal{D}_T$  is the temperature domain of the model and  $f_{\lambda}(\cdot)$  denotes the membrane water-content dynamics in (2). Accordingly, the optimal control problem is formulated as follows:

$$\min_{u(\cdot)} \int_{t_0}^{t_N} P_{\text{VTMS}}(u(t)) dt \quad (10a)$$

subject to

$$\dot{x}(t) = f_x(x(t), u(t), d(t)), \quad (10b)$$

$$\dot{Q}_{\text{cabDuty}}(t) = f_{\text{cabDuty}}(d(t)), \quad (10c)$$

$$\dot{Q}_{\text{WH}}(t) = f_{\text{WH}}(x(t), u(t), d(t)), \quad (10d)$$

$$\dot{Q}_{\text{PTC}}(t) = \max(0, \dot{Q}_{\text{cabDuty}}(t) - \dot{Q}_{\text{WH}}(t)), \quad (10e)$$

$$\underline{x}(d(t)) \leq x(t) \leq \bar{x}(d(t)), \quad (10f)$$

$$\underline{u} \leq u(t) \leq \bar{u}, \quad (10g)$$

where  $f_x(\cdot)$  denotes the thermal dynamics in (1);  $f_{\text{WH}}(\cdot)$  is obtained from the heat-exchanger model in (1c) and determines the waste heat recovered for cabin heating; and  $\dot{Q}_{\text{PTC}}$  supplies any remaining cabin heating demand in (1b). The state bounds combine the water-management temperature bounds with the thermal operating limits as

$$\begin{aligned} \underline{x}(d(t)) &= \begin{bmatrix} \max\{\underline{T}_{\text{st}}, \underline{T}_{\text{st,WM}}(d_{\text{WM}})\} \\ \underline{T}_{\text{PE}} \end{bmatrix}, \\ \bar{x}(d(t)) &= \begin{bmatrix} \min\{\bar{T}_{\text{st}}, \bar{T}_{\text{st,WM}}(d_{\text{WM}})\} \\ \bar{T}_{\text{PE}} \end{bmatrix}, \end{aligned} \quad (11)$$

where  $\underline{T}_i$  and  $\overline{T}_i$  denote the lower and upper thermal operating limits of component  $i \in \{\text{st}, \text{PE}\}$ , respectively. The input bounds  $\underline{u}$  and  $\overline{u}$  specify the allowable heat-transfer rates.

### III. WATER-THERMAL EQUIVALENT CONSUMPTION MINIMIZATION STRATEGY

#### A. WT-ECMS

The proposed water-thermal equivalent consumption minimization strategy (WT-ECMS) coordinates VTMS power consumption and thermal storage under hydration-dependent temperature bounds. Based on Pontryagin's minimum principle, the Hamiltonian is

$$H(x, u, d, \psi) = P_{\text{VTMS}}(u) + \psi^T f_x(x, u, d), \quad (12)$$

where  $\psi$  is the thermal costate vector. The two terms represent the VTMS power consumption and the costate-weighted temperature variation, respectively.

Owing to the thermal inertia of the components, their temperatures vary only slightly over a sufficiently short control interval. The state dependence of the thermal dynamics is therefore locally frozen at the current temperature  $\tilde{x}$ , with the disturbance held at its current value  $\tilde{d}$ . The approximated Hamiltonian becomes

$$\begin{aligned} \tilde{H}(u, \psi) &= P_{\text{VTMS}}(u) + \psi^T \tilde{f}_x(u), \\ \tilde{f}_x(u) &:= f_x(\tilde{x}, u, \tilde{d}). \end{aligned} \quad (13)$$

For this locally approximated problem, the PMP necessary conditions are

$$\dot{x}^* = \frac{\partial \tilde{H}}{\partial \psi} = \tilde{f}_x(u^*), \quad (14a)$$

$$\dot{\psi} = -\frac{\partial \tilde{H}}{\partial x} = 0, \quad (14b)$$

$$0 = \frac{\partial \tilde{H}}{\partial u} = \frac{\partial P_{\text{VTMS}}}{\partial u} + \left( \frac{\partial \tilde{f}_x}{\partial u} \right)^T \psi, \quad (14c)$$

where (14c) applies to an interior minimizer. Equation (14b) implies that the costate remains constant within the control interval. The PMP minimum condition then determines the requested heat-transfer rates as

$$\begin{aligned} u^*(t) &= \arg \min_u \tilde{H}(u, \psi(t)), \\ \text{subject to} \quad & \underline{x}(d(t)) \leq x(t) + \Delta t \tilde{f}_x(u) \leq \overline{x}(d(t)), \quad \underline{u} \leq u \leq \overline{u}. \end{aligned} \quad (15)$$

Since  $x(t)$  cannot be changed instantaneously by  $u$ , the state constraint is imposed on the temperature predicted over the control interval  $\Delta t$ , with the bounds held at their current values.

#### B. WT-AECMS

In WT-ECMS, the costate determines the penalty assigned to temperature variation. Although it remains constant within each control interval under the local approximation, using a fixed value throughout operation assigns the same weight regardless of the thermal condition. A temperature increase

near the upper allowable limit should be penalized more strongly than the same increase near the lower limit.

The proposed WT-AECMS therefore adjusts the costate according to the current temperature and admissible lower bound:

$$\psi(t) = \psi_{\text{base}} + K_p(x(t) - \underline{x}(d(t))), \quad (16)$$

where  $\psi_{\text{base}}$  is the baseline costate vector and  $K_p$  is a positive diagonal gain matrix, both treated as tuning parameters. This adjustment increases the penalty on further temperature rise as the temperature increases relative to the lower bound.

### IV. RESULTS

#### A. Setup

The proposed WT-AECMS is evaluated in a GT-SUITE/Python co-simulation environment. A high-fidelity VTMS model developed in GT-SUITE from real driving data [20] serves as the plant, while the Python controller determines the requested heat-transfer rates at each control interval.

Simulations are conducted over the WLTC at an ambient temperature of  $-10^\circ\text{C}$ . This heating condition requires coordinated thermal storage and waste-heat recovery to reduce auxiliary heating power while respecting hydration-dependent stack temperature bounds, providing a suitable scenario for evaluating the proposed strategy.

#### B. Implementation Details

At cold start, the component temperatures lie below their admissible ranges. For each component below its lower bound, a predefined warm-up policy suppresses heat rejection and operates the dedicated heater, where available, at maximum power, while the PTC heater supplies the cabin heating demand. Once a component temperature enters its admissible range, its control inputs are determined by WT-AECMS at the control interval  $\Delta t = 3$  s. If the temperature later leaves the range, e.g., owing to a change in the hydration-dependent bounds, predefined recovery logic is applied until it re-enters the range, applying the warm-up policy below the range and maximum heat rejection above it, after which WT-AECMS resumes.

The cabin heating demand function  $f_{\text{cabDuty}}(\cdot)$  in (10c) and the thermal operating limits are determined from real driving data of the target vehicle. The temperature limits are

$$\begin{aligned} [\underline{T}_{\text{st}}, \underline{T}_{\text{PE}}]^T &= [55, -5]^T \quad [^\circ\text{C}], \\ [\overline{T}_{\text{st}}, \overline{T}_{\text{PE}}]^T &= [75, 60]^T \quad [^\circ\text{C}], \end{aligned} \quad (17)$$

Here, the stack temperature limits are further restricted by the hydration-dependent bounds defined above. The allowable membrane water-content range is set to  $\underline{\lambda}_{\text{m}} = 10$  and  $\overline{\lambda}_{\text{m}} = 20$ . The baseline costate and gain matrix in (16) are set to  $\psi_{\text{base}} = [-60, 10]^T$  and  $K_p = \text{diag}(4, 2)$ , respectively, with entries corresponding to the stack and PE.

Three control strategies are evaluated: the baseline controller [20], thermal AECMS (T-AECMS), and the proposed

TABLE I: Quantitative performance comparison among the control strategies: WT-AECMS, T-AECMS, and Baseline.

| Method   | Energy Usage (kWh) ↓ | ATTR <sub>st</sub> ↑ | ATTR <sub>PE</sub> ↑ | AHTR ↑      |
|----------|----------------------|----------------------|----------------------|-------------|
| WT-AECMS | 1.35                 | <b>0.94</b>          | <b>1.00</b>          | <b>1.00</b> |
| T-AECMS  | <b>1.32</b>          | 0.92                 | 1.00                 | 0.70        |
| Baseline | 3.45                 | 0.91                 | 0.90                 | 0.91        |

WT-AECMS. The baseline implements the production control logic used in the target vehicle, combining rule-based operation with temperature reference tracking. T-AECMS optimizes system-level heat allocation using adaptive costates under thermal operating limits, without incorporating hydration constraints. It uses the same baseline costate vector and adaptation gain matrix as WT-AECMS. WT-AECMS extends this formulation by introducing hydration-dependent stack temperature bounds.

The proposed strategy is evaluated based on VTMS energy consumption and thermal and hydration constraint satisfaction. Total VTMS energy consumption quantifies the electrical energy required for thermal management. Constraint satisfaction is assessed using the allowable temperature time ratio (ATTR) and the allowable hydration time ratio (AHTR), defined as

$$\text{ATTR}_i = \frac{1}{t_N - t_0} \int_{t_0}^{t_N} \mathbf{1}[x_i(t) \leq x_i(t) \leq \bar{x}_i(t)] dt, \quad (18)$$

$$\text{AHTR} = \frac{1}{t_N - t_0} \int_{t_0}^{t_N} \mathbf{1}[\lambda_m \leq \lambda_m(t) \leq \bar{\lambda}_m] dt,$$

where  $\mathbf{1}[\cdot]$  is the indicator function,  $x_i$  is the temperature of component  $i$ , and  $\lambda_m$  is the membrane water content. The temperature bounds include the hydration-dependent limits for the stack, while  $\lambda_m$  and  $\bar{\lambda}_m$  denote the allowable membrane water-content bounds. Each ratio represents the fraction of the driving scenario during which the corresponding constraint is satisfied, with unity indicating satisfaction throughout the scenario. The ATTR is evaluated over the entire scenario, including the initial warm-up and any subsequent out-of-range periods.

### C. Results

Table I compares the three strategies. T-AECMS reduces the total VTMS energy consumption from 3.45 kWh to 1.32 kWh, approximately 62% lower than the baseline, and raises ATTR<sub>PE</sub> from 0.90 to 1.00 while slightly improving ATTR<sub>st</sub> from 0.91 to 0.92. However, its AHTR falls to 0.70, compared with 0.91 for the baseline, indicating that satisfying the thermal limits alone does not ensure adequate membrane hydration.

WT-AECMS addresses this issue by incorporating the hydration-dependent temperature bounds. It achieves an AHTR of 1.00, raises ATTR<sub>st</sub> to 0.94, and maintains ATTR<sub>PE</sub> at 1.00. Its energy consumption is 1.35 kWh, only 0.03 kWh, or approximately 2.3%, higher than that of T-AECMS and approximately 61% lower than that of

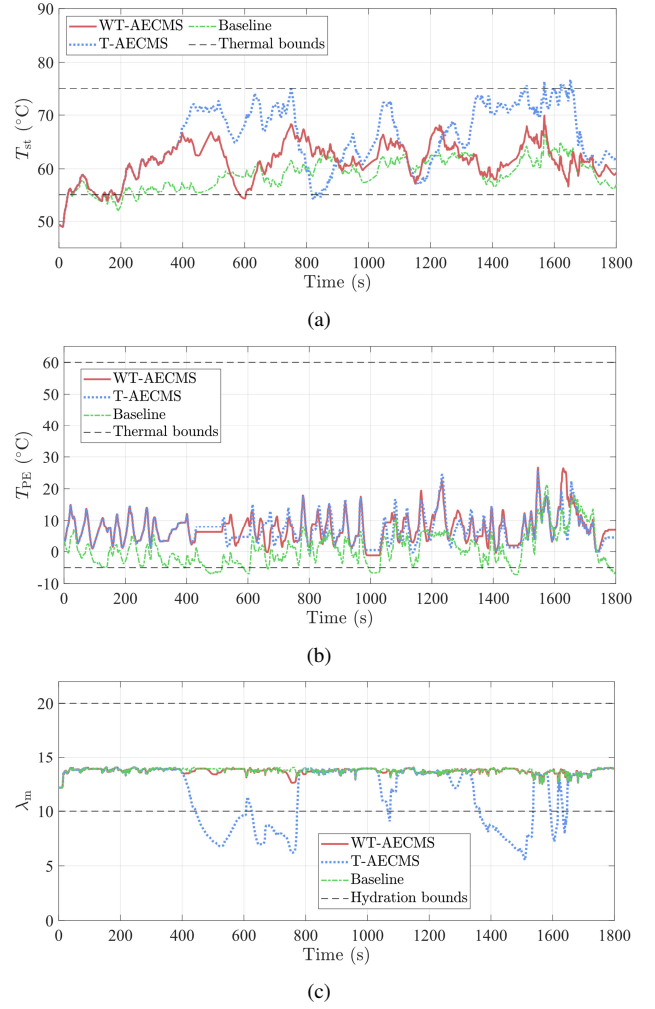


Fig. 1: Comparison of trajectories under WT-AECMS, T-AECMS, and the baseline controller: (a) stack temperature, (b) PE temperature, and (c) membrane water content. Dashed gray lines indicate the corresponding operating limits.

the baseline. Incorporating water-management constraints therefore preserves most of the energy savings of thermal optimization while keeping the membrane water content within its allowable range.

Figs. 1(a)–(c) show the stack temperature, PE temperature, and membrane water content, respectively, with gray dashed lines indicating the limits. After the initial warm-up, the baseline keeps the stack temperature mostly between 55 and 65 °C. T-AECMS exploits a wider range to store waste heat, repeatedly exceeding 70 °C and slightly violating the upper limit near 1600 s. WT-AECMS also uses the stack as a thermal buffer but at lower temperatures, consistent with the hydration-dependent upper bound. For the PE, both AECMS strategies remain above the lower limit throughout, whereas the baseline falls below  $-5$  °C at several points; their PE trajectories nearly coincide since the hydration-dependent bounds apply only to the stack.

The effect on hydration is shown in Fig. 1(c). Under T-AECMS, the membrane water content drops below the lower limit during periods of elevated stack temperature,

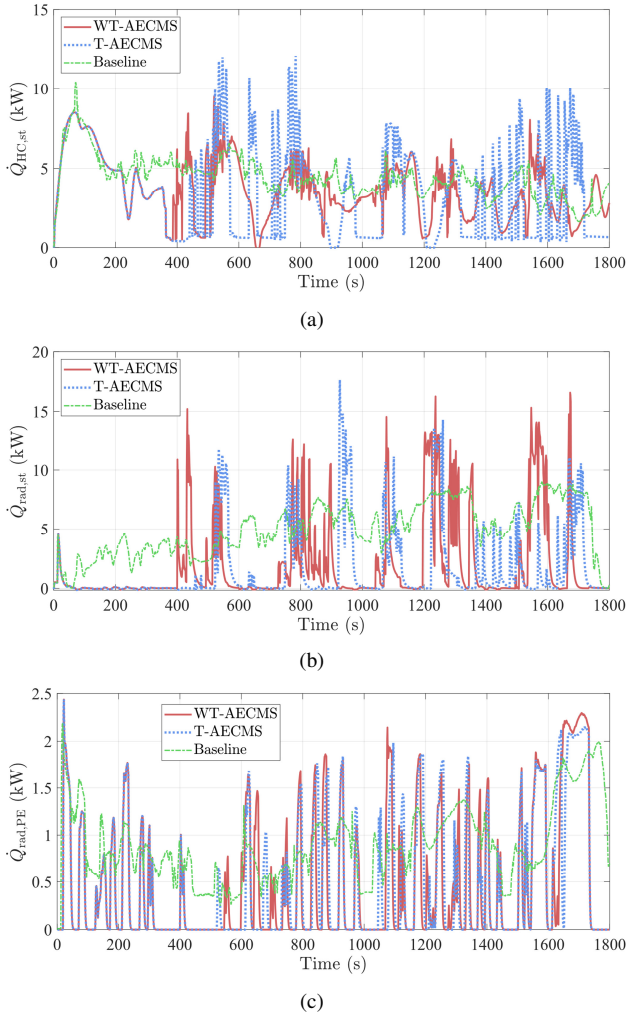


Fig. 2: Comparison of heat-transfer rates under WT-AECMS, T-AECMS, and the baseline controller: (a) stack heater core, (b) stack radiator, and (c) PE radiator.

whereas WT-AECMS maintains it within the allowable range throughout the scenario. These trajectories confirm the coupling between stack temperature and hydration: satisfying the thermal limits alone can still permit membrane drying, which the hydration-dependent bounds prevent.

Fig. 2 compares the corresponding heat-transfer rates. The baseline rejects stack heat continuously through the radiator, whereas both AECMS strategies retain heat for longer periods and reject it in short bursts of up to about 15 kW. Compared with T-AECMS, WT-AECMS activates the stack radiator during intervals in which T-AECMS continues to retain heat, e.g., around 400–500 s and 1500–1600 s, limiting the stack temperature rise in Fig. 1(a), and its heater-core heat-transfer rate fluctuates less. The PE radiator trajectories of the two AECMS strategies nearly coincide, consistent with their PE temperatures.

Fig. 3 shows the breakdown of the VTMS energy consumption. Both AECMS strategies substantially reduce the component-heater energy relative to the baseline, which accounts for most of the overall savings despite an increase

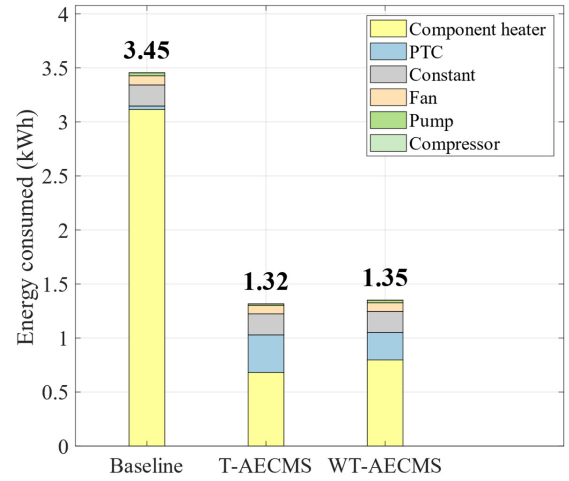


Fig. 3: Total VTMS energy consumption and its breakdown under the baseline controller, T-AECMS, and WT-AECMS.

in PTC heater consumption. Compared with T-AECMS, WT-AECMS consumes slightly more energy for component heating but less for the PTC heater. These opposing changes largely offset each other, resulting in a net increase of only 0.03 kWh; incorporating the hydration constraints therefore has a limited effect on the total VTMS energy consumption.

Fig. 4 shows the thermal-energy flows, with the module-level VTMS energy consumption given in each box. Under the baseline, the stack module consumes 11316 kJ while 8901 kJ of stack heat is rejected through the underhood module; although the baseline recovers more heat for cabin heating, this combination of electric heating and heat rejection results in high total consumption. T-AECMS and WT-AECMS reduce stack heat rejection to 2930 kJ and 3385 kJ and stack-module consumption to 2494 kJ and 2948 kJ, respectively. Their lower heat recovery increases cabin heating consumption, but this increase is far smaller than the reduction in stack-module consumption. Compared with T-AECMS, WT-AECMS removes more heat from the stack through both the radiator and cabin heat recovery, consistent with its hydration-dependent temperature limits. The recovered heat increases from 6155 kJ to 6635 kJ, reducing cabin-module consumption from 1302 kJ to 969 kJ, which largely offsets the 454 kJ increase in stack-module energy.

## V. CONCLUSION

This study proposed WT-AECMS, a PMP-based strategy for FCEV water-thermal management that uses heat-transfer rates as control inputs and incorporates hydration-dependent stack temperature bounds, with costates adapted to each component temperature relative to its admissible range. In high-fidelity co-simulations over the WLTC at  $-10^\circ\text{C}$ , WT-AECMS reduced VTMS energy consumption by approximately 61% relative to the baseline while improving both thermal and hydration constraint satisfaction. Compared with T-AECMS, it increased AHTR from 0.70 to 1.00 at the cost of only approximately 2% additional energy. These results

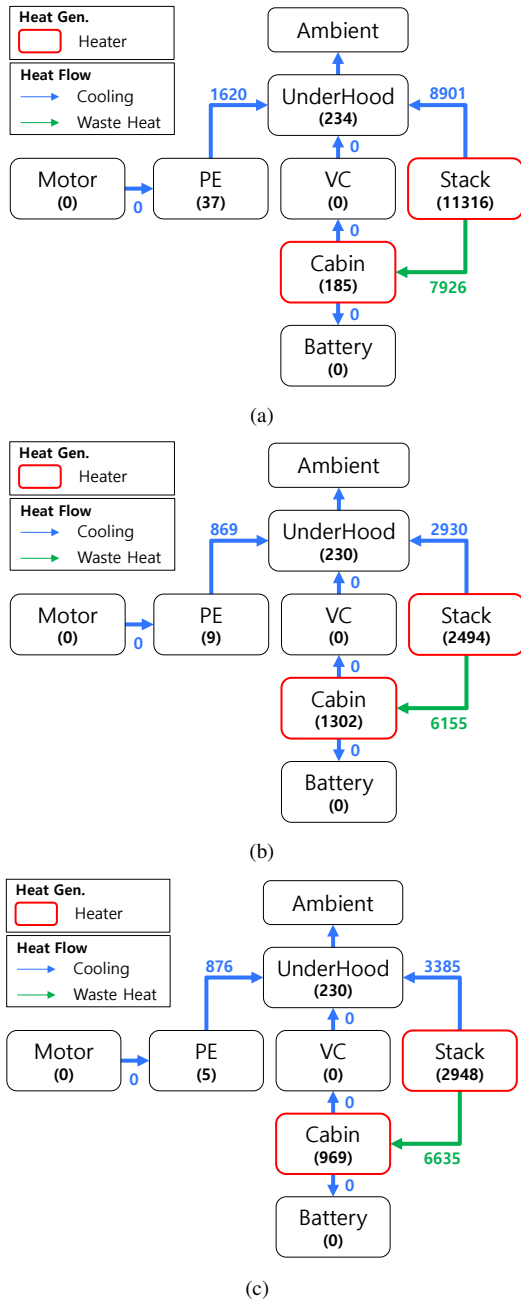


Fig. 4: Thermal-energy flow distributions under (a) the baseline controller, (b) T-AECMS, and (c) WT-AECMS.

indicate that thermal limits alone do not ensure adequate membrane hydration and that hydration should be accounted for in vehicle-level heat allocation. Future work will examine robustness across driving cycles and ambient conditions and validate the strategy experimentally.

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